

Algebra Sparks

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Find the Rule

Say the rule out loud in one sentence before you write the next number.

Warm-Up 10 ITEMS • RECALL

1 3, 6, 9, 12 — next

2 5, 10, 15, 20 — next

3 2, 4, 8, 16 — next

4 20, 17, 14, 11 — next

5 1, 4, 7, 10 — next

6 100, 90, 80 — next

7 2, 4, 6, 8 — next

8 10, 20, 30 — next

9 1, 3, 5, 7 — next

10 4, 8, 12 — next

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

11. Rule $\times 3$, starting at 2 — the fourth number (2, 6, 18, 54)

.....

12. 1, 3, 7, 15 — next (Double it and add one)

.....

13. Rule $\times 2 + 1$. Put in 5

.....

14. 2, 5, 10, 17 — next (The gaps go 3, 5, 7, 9)

.....

15. 7, 14, 21, 28 — next

.....

16. 1, 4, 9, 16 — next

.....

17. 3, 9, 27 — next

.....

Function Machines

In goes a number, out comes a number. The machine never changes its mind halfway.

Warm-Up 12 ITEMS • RECALL

1 Machine + 8. In 5

2 Machine $\times$ 6. In 4

3 Machine $-$ 7. In 20

4 Machine $\times$ 10. In 9

5 Machine $\div$ 2. In 18

6 Machine + 15. In 15

7 Machine $\times$ 2. In 5

8 Machine $\times$ 3. In 4

9 Machine + 7. In 8

10 Machine $-$ 3. In 10

11 Machine $\times$ 10. In 6

12 Machine $\div$ 2. In 14

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Machine $\times$ 3 + 4. In 6 (Multiply first)

.....

14. Machine $\times$ 5 $-$ 2. In 7

.....

15. Machine $\times$ 2 + 9. In 12

.....

16. Machine $\times$ 3 + 4. Out 25 — what went in (Undo the + 4 first)

.....

17. Machine $\times$ 4 $-$ 1. Out 27 — what went in

.....

18. Machine $\times$ 3 + 1. In 7

.....

19. Machine $\times$ 4 $-$ 2. In 6

.....

20. Machine $\times$ 5 + 3. In 9

.....

21. Machine $\times$ 2 + 10. In 12

.....

22. Machine $\times$ 6 $-$ 5. In 8

.....

★ Challenge NOT GRADED • REASON IT OUT

23. In 2 gives 7, in 3 gives 10, in 4 gives 13. In 10 gives (The rule is $\times 3 + 1$)

24. In 1 gives 5, in 2 gives 9, in 3 gives 13. In 12 gives (The rule is $\times 4 + 1$)

25. Machine $\times 3 + 4$ gives 31 — what went in

26. Machine $\times 7 - 2$ gives 47 — what went in

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Letters Standing for Numbers

A letter is a number nobody has named yet. Put the number in, then work it out.

Warm-Up 12 ITEMS • RECALL

1 $n = 4$. Value of $n + 6$

2 $n = 4$. Value of $3n$

3 $n = 7$. Value of $n - 3$

4 $n = 5$. Value of $2n$

5 $n = 9$. Value of $n + n$

6 $n = 6$. Value of $10 - n$

7 $n = 3$. Value of $n + 5$

8 $n = 3$. Value of $4n$

9 $n = 8$. Value of $n - 5$

10 $n = 6$. Value of $2n$

11 $n = 7$. Value of $n + n$

12 $n = 4$. Value of $12 - n$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $n = 5$. Value of $3n + 2$

.....

14. $n = 8$. Value of $2n - 5$

.....

15. $a = 3$, $b = 7$. Value of $4a + b$ (12 and 7)

.....

16. $n = 6$. Value of $n \times n$

.....

17. $n = 12$. Value of $n \div 4 + 5$

.....

18. $n = 6$. Value of $3n + 4$

.....

19. $n = 9$. Value of $2n - 7$

.....

20. $n = 5$. Value of $n \times n$

.....

21. $n = 20$. Value of $n \div 4 + 3$

.....

22. $n = 11$. Value of $4n - 14$

.....



Working space



Error journal

One-Step Equations

Do the same thing to both sides. Type the value of the letter.

Warm-Up 12 ITEMS • RECALL

1 $x + 4 = 11$

2 $x - 5 = 9$

3 $3x = 21$

4 $x + 12 = 20$

5 $6x = 30$

6 $x \div 2 = 8$

7 $x + 3 = 10$

8 $x + 6 = 14$

9 $x + 9 = 15$

10 $x + 4 = 4$

11 $x + 11 = 20$

12 $x + 2 = 13$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $x + 17 = 42$

14. $8x = 96$ ($96 \div 8$)

15. $x - 23 = 19$

16. $x \div 5 = 13$

17. $4x = 100$

18. $x + 24 = 51$

19. $x - 7 = 12$

20. $x - 31 = 24$

21. $x + 47 = 100$

22. $x - 18 = 33$

Guess My Rule & the Math Trail

Capstone day. Every stop on the trail is a problem you wrote yourself.

Warm-Up 4 ITEMS • RECALL

1 Stops on a ten-stop trail

2 $x + 9 = 9$

3 $3x = 12$

4 $5x = 20$

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Trail rule $\times 2 + 3$. Stop number 6 gives

.....

6. Ten stops, 3 minutes each — total minutes

.....

7. $9x = 72$

.....

8. $x \div 4 = 11$

.....

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Find the Rule

OUT OF 17

Warm-Up • 1-10 • 15 | 25 | 32 | 8 | 13 | 70 | 10 | 40 | 9 | 16

Core • 11-17 • 54 | 31 | 11 | 26 | 35 | 25 | 81

Challenge • 18-22 • 49 | 25 | 101 | 13 | 42

1.2 • Function Machines

OUT OF 22

Warm-Up • 1-12 • 13 | 24 | 13 | 90 | 9 | 30 | 10 | 12 | 15 | 7 | 60 | 7

Core • 13-22 • 22 | 33 | 33 | 7 | 7 | 22 | 22 | 48 | 34 | 43

Challenge • 23-26 • 31 | 49 | 9 | 7

1.3 • Letters Standing for Numbers

OUT OF 22

Warm-Up • 1-12 • 10 | 12 | 4 | 10 | 18 | 4 | 8 | 12 | 3 | 12 | 14 | 8

Core • 13-22 • 17 | 11 | 19 | 36 | 8 | 22 | 11 | 25 | 8 | 30

Challenge • 23-26 • 20 | 28 | 36 | 37

1.4 • One-Step Equations

OUT OF 22

Warm-Up • 1-12 • 7 | 14 | 7 | 8 | 5 | 16 | 7 | 8 | 6 | 0 | 9 | 11

Core • 13-22 • 25 | 12 | 42 | 65 | 25 | 27 | 19 | 55 | 53 | 51

Challenge • 23-26 • 7 | 11 | 266 | 200

Fri • Guess My Rule & the Math Trail

OUT OF 8

Warm-Up • 1-4 • 10 | 0 | 4 | 4

Core • 5-8 • 15 | 30 | 8 | 44

Challenge • 9-14 • 43 | 10 | 16 | 23 | 9 | 9

What a Letter Is

n is a seat a number sits in. $3n$ is three of them.

Warm-Up 12 ITEMS • RECALL

1 $n = 4$. Value of $n + 6$

2 $n = 4$. Value of $3n$

3 $n = 7$. Value of $n - 3$

4 $n = 5$. Value of $2n$

5 $n = 9$. Value of $n + n$

6 $n = 6$. Value of $10 - n$

7 $n = 5$. Value of $n + 8$

8 $n = 5$. Value of $4n$

9 $n = 9$. Value of $n - 4$

10 $n = 7$. Value of $3n$

11 $n = 8$. Value of $n + n$

12 $n = 3$. Value of $15 - n$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $n = 5$. Value of $3n + 2$

.....

14. $n = 8$. Value of $2n - 5$

.....

15. $n = 6$. Value of $n \times n$

.....

16. $n = 12$. Value of $n \div 4 + 5$

.....

17. $n = 10$. Value of $4n - 10$

.....

18. $n = 7$. Value of $4n + 5$

.....

19. $n = 9$. Value of $3n - 8$

.....

20. $n = 8$. Value of $n \times n$

.....

21. $n = 24$. Value of $n \div 6 + 7$

.....

22. $n = 15$. Value of $3n - 20$

.....



Working space



Error journal

Evaluate It

Substitute the number, then work it out. One step at a time.

Warm-Up 12 ITEMS • RECALL

1 $n = 2$. Value of $5n$

2 $n = 3$. Value of $n + 7$

3 $n = 10$. Value of $n \div 2$

4 $n = 1$. Value of $9n$

5 $n = 0$. Value of $6n$

6 $n = 8$. Value of $n - 8$

7 $n = 3$. Value of $7n$

8 $n = 4$. Value of $n + 11$

9 $n = 12$. Value of $n \div 3$

10 $n = 2$. Value of $8n$

11 $n = 0$. Value of $9n$

12 $n = 6$. Value of $n - 6$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $n = 7$. Value of $4n + 1$
.....

14. $n = 6$. Value of $5n - 12$
.....

15. $n = 9$. Value of $2n + 2$
.....

16. $n = 11$. Value of $3n - 3$
.....

17. $n = 4$. Value of $25 - 3n$
.....

18. $n = 8$. Value of $5n + 2$
.....

19. $n = 7$. Value of $6n - 10$
.....

20. $n = 10$. Value of $3n + 5$
.....

21. $n = 12$. Value of $4n - 8$
.....

22. $n = 5$. Value of $40 - 4n$
.....

Write the Expression

Turn the sentence into symbols. Then check it on a number.

Warm-Up 12 ITEMS • RECALL

1 Three lots of n , when $n = 4$

2 n more than 5, when $n = 3$

3 Double n , when $n = 7$

4 n less than 10, when $n = 4$

5 Half of n , when $n = 8$

6 n plus n , when $n = 5$

7 Four lots of n , when $n = 5$

8 n more than 8, when $n = 4$

9 Double n , when $n = 9$

10 n less than 20, when $n = 7$

11 Half of n , when $n = 12$

12 n plus n , when $n = 11$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Five more than double n , when $n = 6$

14. Three less than four lots of n , when $n = 5$

15. Ten more than half of n , when $n = 20$

16. Triple n , plus 4, when $n = 8$

17. n multiplied by itself, when $n = 7$

18. Seven more than double n , when $n = 8$

19. Four less than five lots of n , when $n = 6$

20. Twelve more than half of n , when $n = 30$

21. Quadruple n , plus 6, when $n = 7$

22. n multiplied by itself, when $n = 9$



Working space



Error journal

Two Letters

Substitute both before you simplify anything.

Warm-Up 12 ITEMS • RECALL

1 $a = 2, b = 3$. Value of $a + b$

2 $a = 2, b = 3$. Value of ab

3 $a = 5, b = 1$. Value of $a - b$

4 $a = 4, b = 4$. Value of $a + b$

5 $a = 3, b = 6$. Value of $b \div a$

6 $a = 1, b = 9$. Value of ab

7 $a = 3, b = 4$. Value of $a + b$

8 $a = 3, b = 4$. Value of ab

9 $a = 9, b = 2$. Value of $a - b$

10 $a = 5, b = 5$. Value of $a + b$

11 $a = 4, b = 12$. Value of $b \div a$

12 $a = 1, b = 11$. Value of ab

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $a = 3, b = 7$. Value of $4a + b$

14. $a = 4, b = 6$. Value of $3a + 2b$

15. $a = 5, b = 2$. Value of $a \times b + 3$

16. $a = 6, b = 3$. Value of $2a - b$

17. $a = 8, b = 2$. Value of $a \div b + 5$

18. $a = 5, b = 6$. Value of $4a + b$

19. $a = 3, b = 8$. Value of $3a + 2b$

20. $a = 7, b = 3$. Value of $a \times b + 4$

21. $a = 9, b = 5$. Value of $2a - b$

22. $a = 12, b = 3$. Value of $a \div b + 6$

Perimeter as a Rule

Write it as $4s$ once, then test it on three squares.

Warm-Up 4 ITEMS • RECALL

1 Perimeter of a square of side 3

2 Of side 5

3 Perimeter of a square of side 4

4 Of side 8

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Using $4s$ with $s = 9$ — the perimeter

.....

6. A square of perimeter 28 — its side

.....

7. Using $4s$ with $s = 11$ — the perimeter

.....

8. A square of perimeter 44 — its side

.....

★ Challenge NOT GRADED • REASON IT OUT

9. Using $2l + 2w$ with $l = 8$, $w = 5$

10. A rectangle of perimeter 30 and length 10 — its width

11. Using $4s$, the side when the perimeter is 100

12. Using $2l + 2w$ with $l = 12$, $w = 7$

13. A rectangle of perimeter 40 and length 14 — its width

14. Using $4s$, the side when the perimeter is 144

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 2 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

2.1 • What a Letter Is

OUT OF 22

Warm-Up • 1-12 • 10 | 12 | 4 | 10 | 18 | 4 | 13 | 20 | 5 | 21 | 16 | 12

Core • 13-22 • 17 | 11 | 36 | 8 | 30 | 33 | 19 | 64 | 11 | 25

Challenge • 23-26 • 20 | 28 | 35 | 24

2.2 • Evaluate It

OUT OF 22

Warm-Up • 1-12 • 10 | 10 | 5 | 9 | 0 | 0 | 21 | 15 | 4 | 16 | 0 | 0

Core • 13-22 • 29 | 18 | 20 | 30 | 13 | 42 | 32 | 35 | 40 | 20

Challenge • 23-26 • 8 | 30 | 6 | 42

2.3 • Write the Expression

OUT OF 22

Warm-Up • 1-12 • 12 | 8 | 14 | 6 | 4 | 10 | 20 | 12 | 18 | 13 | 6 | 22

Core • 13-22 • 17 | 17 | 20 | 28 | 49 | 23 | 26 | 27 | 34 | 81

Challenge • 23-26 • 16 | 28 | 30 | 48

2.4 • Two Letters

OUT OF 22

Warm-Up • 1-12 • 5 | 6 | 4 | 8 | 2 | 9 | 7 | 12 | 7 | 10 | 3 | 11

Core • 13-22 • 19 | 24 | 13 | 9 | 9 | 26 | 25 | 25 | 13 | 10

Challenge • 23-26 • 13 | 20 | 12 | 26

Fri • Perimeter as a Rule

OUT OF 8

Warm-Up • 1-4 • 12 | 20 | 16 | 32

Core • 5-8 • 36 | 7 | 44 | 11

Challenge • 9-14 • 26 | 5 | 25 | 38 | 6 | 36

Keep It Level

Take the same amount from both sides and the balance holds.

✦ Warm-Up 12 ITEMS • RECALL

$$1 \quad x + 4 = 11$$

$$2 \quad x + 5 = 12$$

$$3 \quad x + 10 = 15$$

$$4 \quad x + 2 = 9$$

$$5 \quad x + 7 = 7$$

$$6 \quad x + 12 = 20$$

$$7 \quad x + 6 = 13$$

$$8 \quad x + 8 = 15$$

$$9 \quad x + 12 = 18$$

$$10 \quad x + 3 = 11$$

$$11 \quad x + 9 = 9$$

$$12 \quad x + 15 = 25$$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

$$13. \quad x + 17 = 42$$

$$14. \quad x - 5 = 9$$

$$15. \quad x - 23 = 19$$

$$16. \quad x + 38 = 100$$

$$17. \quad x - 14 = 27$$

$$18. \quad x + 29 = 55$$

$$19. \quad x - 8 = 14$$

$$20. \quad x - 35 = 27$$

$$21. \quad x + 56 = 120$$

$$22. \quad x - 19 = 42$$

Multiply and Divide

Divide both sides by the same number and the balance still holds.

Warm-Up 12 ITEMS • RECALL

1 $3x = 21$

2 $6x = 30$

3 $2x = 14$

4 $5x = 25$

5 $x \div 2 = 8$

6 $4x = 20$

7 $4x = 24$

8 $7x = 35$

9 $3x = 18$

10 $9x = 45$

11 $x \div 3 = 7$

12 $5x = 40$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $8x = 96$

14. $7x = 84$

15. $x \div 5 = 13$

16. $4x = 100$

17. $9x = 108$

18. $6x = 90$

19. $8x = 120$

20. $x \div 6 = 14$

21. $11x = 132$

22. $12x = 156$



Check Your Answer

Put it back in. If both sides match, you are done — no marking needed.

Warm-Up 12 ITEMS • RECALL

1 $x = 7$ in $x + 4$ — the value

2 $x = 5$ in $3x$ — the value

3 $x = 9$ in $x - 2$

4 $x = 4$ in $2x + 1$

5 $x = 6$ in $x \div 2$

6 $x = 10$ in $x + 10$

7 Every answer in the 4 times table is a multiple of what

8 Every answer in the 6 times table is a multiple of what

9 Every answer in the 8 times table is a multiple of what

10 Every answer in the 9 times table is a multiple of what

11 2×5 — does it end in 0 or 5

12 4×5 — does it end in 0 or 5

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

22. Solve then check: $3x = 27$

23. Check $x = 9$ in $3x$ — the value

24. Solve then check: $x + 19 = 31$

25. Check $x = 12$ in $x + 19$

26. Solve: $6x = 72$

27. 3×5 — does it end in 0 or 5

28. 5×5 — does it end in 0 or 5

29. 7×5 — does it end in 0 or 5

30. 9×5 — does it end in 0 or 5

31. Adding two odd numbers gives an odd or an even answer

32. Adding an odd and an even number gives

★ Challenge NOT GRADED • REASON IT OUT

42. $2x + 5 = 19$

43. Check $x = 7$ in $2x + 5$

44. In the multiplication table, the diagonal is made of what kind of number

45. $3x + 4 = 25$

46. Check $x = 7$ in $3x + 4$



Working space



Error journal

Word to Equation

Turn the situation into an equation, then solve it.

Warm-Up 12 ITEMS • RECALL

1 A number plus 5 is 12.
The number

2 Double a number is 14.
The number

3 A number less 3 is 8. The
number

4 Three lots of a number is
18. The number

5 Half a number is 9. The
number

6 A number plus itself is 16.
The number

7 A number plus 8 is 15.
The number

8 Double a number is 18.
The number

9 A number less 5 is 12.
The number

10 Four lots of a number is
24. The number

11 Half a number is 11. The
number

12 A number plus itself is
24. The number

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Four friends share equally and get 13 each. The total

.....

14. After spending \$18 you have \$27. You started with

.....

15. Five boxes hold 60 altogether. Each box

.....

16. A number times 7 is 91. The number

.....

17. Twice a number plus 3 is 21. The number

.....

18. Five friends share equally and get 14 each. The total

.....

19. After spending \$23 you have \$34. You started with

.....

20. Six boxes hold 90 altogether. Each box

.....

21. A number times 8 is 96. The number

.....

22. Twice a number plus 5 is 27. The number

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

Warm-Up 4 ITEMS • RECALL

1 6, 11, 16, 21 — next

2 $n = 4$. Value of $3n$

3 4, 9, 14, 19 — next

4 $n = 5$. Value of $4n$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Machine $\times 4 + 2$. In 9

.....

6. That machine gives 30 — what went in

.....

7. $n = 7$. Value of $3n - 4$

.....

8. $x + 18 = 45$

.....

9. $7x = 63$

.....

10. Machine $\times 5 + 3$. In 8

.....

11. That machine gives 28 — what went in

.....

12. $n = 8$. Value of $4n - 6$

.....

13. $x + 26 = 51$

.....

14. $8x = 72$

.....

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Keep It Level

OUT OF 22

Warm-Up • 1-12 • 7 | 7 | 5 | 7 | 0 | 8 | 7 | 7 | 6 | 8 | 0 | 10

Core • 13-22 • 25 | 14 | 42 | 62 | 41 | 26 | 22 | 62 | 64 | 61

Challenge • 23-26 • 244 | 200 | 312 | 300

3.2 • Multiply and Divide

OUT OF 22

Warm-Up • 1-12 • 7 | 5 | 7 | 5 | 16 | 5 | 6 | 5 | 6 | 5 | 21 | 8

Core • 13-22 • 12 | 12 | 65 | 25 | 12 | 15 | 15 | 84 | 12 | 13

Challenge • 23-26 • 24 | 147 | 25 | 171

3.3 • Check Your Answer

OUT OF 23

Warm-Up • 1-12 • 11 | 15 | 7 | 9 | 3 | 20 | 4 | 6 | 8 | 9 | 0 | 0

Core • 22-32 • 9 | 27 | 12 | 31 | 12 | 5 | 5 | 5 | 5 | even | odd

Challenge • 42-46 • 7 | 19 | square | 7 | 25

3.4 • Word to Equation

OUT OF 22

Warm-Up • 1-12 • 7 | 7 | 11 | 6 | 18 | 8 | 7 | 9 | 17 | 6 | 22 | 12

Core • 13-22 • 52 | 45 | 12 | 13 | 9 | 70 | 57 | 15 | 12 | 11

Challenge • 23-26 • 11 | 12 | 10 | 13

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 26 | 12 | 24 | 20

Core • 5-14 • 38 | 7 | 17 | 27 | 9 | 43 | 5 | 26 | 25 | 9

Challenge • 15-16 • 39 | 59

Design the Trail

Ten stops, one per mission plus two of your own. Map it first.

Warm-Up 9 ITEMS • RECALL

1 Stops on the trail

2 Missions in the year

3 Stops of your own choosing

4 3 minutes per stop, 10 stops — minutes

5 Half of 10 stops

6 Two stops per room — rooms needed

7 4 minutes per stop, 10 stops — minutes

8 Half of 8 missions

9 Two stops per station — stations needed

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. Trail rule $\times 2 + 3$ — stop 6 gives

.....

11. That rule — stop 10 gives

.....

12. 10 stops at 4 minutes each — minutes

.....

13. If 3 stops are outdoors, the percent indoors

.....

14. Stops numbered 1 to 10 — their total

.....

15. Trail rule $\times 3 + 2$ — stop 5 gives

.....

16. Trail rule $\times 3 + 2$ — stop 10 gives

.....

17. 10 stops at 5 minutes each — minutes

.....

18. If 4 stops are outdoors, the percent indoors

.....

19. Stops numbered 1 to 12 — their total

.....



Working space



Error journal

Write the Problems

Every stop gets a problem and a hidden answer key in your handwriting.

Warm-Up 8 ITEMS • RECALL

1 23×14

2 $372 \div 3$

3 $2 + 3 \times 4$

4 Perimeter of a 4 by 3 rectangle

5 34×12

6 $484 \div 4$

7 $3 + 4 \times 5$

8 Perimeter of a 5 by 4 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. $x + 17 = 42$

10. Area 48, one side 6 — the other
_____11. GCF of 24 and 36
_____12. Round 4,829 to the nearest thousand

13. $x + 24 = 51$

14. Area 72, one side 8 — the other
_____15. GCF of 18 and 24
_____16. Round 5,632 to the nearest thousand

Challenge NOT GRADED • REASON IT OUT

17. $\frac{1}{2} + \frac{1}{4}$ — type as a/b

18. $\$20 - \13.68

19. Mean of 4, 7, 7, 9, 13

20. $\frac{2}{3} + \frac{3}{4}$ — type as a/b

21. A machine $\times 4 + 3 -$ in 10 gives

22. $\frac{1}{2} + \frac{1}{3}$ — type as a/b

23. \$20 - \$11.35

24. Mean of 6, 8, 8, 10, 13

25. $\frac{3}{4} + \frac{2}{3}$ — type as a/b

26. A machine $\times 5 + 2 \rightarrow$ in 12 gives

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Mom Walks It

You mark her work. Any ambiguous question gets rewritten on the spot.

Warm-Up 12 ITEMS • RECALL

1 10 stops, 8 correct — the percent

2 10 stops, 9 correct — the percent

3 10 stops, 10 correct — the percent

4 10 minus 7

5 Half of 10 stops

6 10 stops, 5 correct — the percent

7 10 stops, 7 correct — the percent

8 10 stops, 6 correct — the percent

9 10 stops, 3 correct — the percent

10 10 minus 4

11 A quarter of 20 stops

12 10 stops, 2 correct — the percent

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. She takes 4 minutes a stop — total minutes
.....

14. Two stops rewritten out of 10 — the percent
.....

15. 85% of 10 stops — how many
.....

16. If 9 of 10 are solvable, the percent that are not
.....

17. She takes 5 minutes a stop — total minutes
.....

18. Three stops rewritten out of 10 — the percent
.....

19. 90% of 10 stops — how many
.....

20. If 8 of 10 are solvable, the percent that are not
.....

Year-End Sweep

All eight error journals. Name the one habit that fixed itself.

Warm-Up 8 ITEMS • RECALL

1 7×8

2 $84 \div 12$

3 $2 + 3 \times 4$

4 $1/2$ of 20

5 6×9

6 $96 \div 12$

7 $3 + 4 \times 5$

8 $1/3$ of 30

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. 38×24

10. $618 \div 6$

11. $3x = 21$

12. 47×23

13. $728 \div 8$

14. $4x = 32$



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Mission 08 Test

Twelve items plus the completed trail. Final trophy band awarded.

Warm-Up 4 ITEMS • RECALL

1 3, 6, 9, 12 — next
_____2 $n = 5$. Value of $2n$
_____3 5, 10, 15, 20 — next
_____4 $n = 6$. Value of $3n$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.5. 2, 6, 18, 54 — next
.....6. A pattern from 5 adding 6 — its twelfth number
.....7. Machine $\times 5 + 3$. In 8
.....8. Machine $\times 6 - 4$ gives 32 — what went in
.....9. $n = 9$. Value of $4n + 7$
.....10. $a = 4$, $b = 6$. Value of $3a + 2b$
.....11. $x - 26 = 19$
.....12. $9x = 108$
.....13. 3, 9, 27, 81 — next
.....14. Machine $\times 6 + 2$. In 7
.....15. Machine $\times 5 - 3$ gives 32 — what went in
.....16. $n = 8$. Value of $5n + 6$
.....17. $a = 5$, $b = 7$. Value of $3a + 2b$
.....

Challenge NOT GRADED • REASON IT OUT

20. $2x + 5 = 19$

21. A machine turns 5 into 23 and 8 into 35 — in 10 gives

22. A pattern from 4 adding 7 — its tenth number

23. $3x + 4 = 25$

24. A machine turns 4 into 19 and 7 into 34 — in 10 gives

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Design the Trail

OUT OF 19

Warm-Up • 1-9 • 10 | 8 | 2 | 30 | 5 | 5 | 40 | 4 | 5

Core • 10-19 • 15 | 23 | 40 | 70 | 55 | 17 | 32 | 50 | 60 | 78

Challenge • 20-23 • 37 | 120 | 46 | 150

4.2 • Write the Problems

OUT OF 16

Warm-Up • 1-8 • 322 | 124 | 14 | 14 | 408 | 121 | 23 | 18

Core • 9-16 • 25 | 8 | 12 | 5000 | 27 | 9 | 6 | 6000

Challenge • 17-26 • $\frac{3}{4}$ | 6.32 | 8 | $\frac{17}{12}$ | 43 | $\frac{5}{6}$ | 8.65 | 9 | $\frac{17}{12}$ | 62

4.3 • Mom Walks It

OUT OF 20

Warm-Up • 1-12 • 80 | 90 | 100 | 3 | 5 | 50 | 70 | 60 | 30 | 6 | 5 | 20

Core • 13-20 • 40 | 20 | 8.5 | 10 | 50 | 30 | 9 | 20

Challenge • 21-26 • 4 | 80 | 102 | 5 | 80 | 127.5

Thu • Year-End Sweep

OUT OF 22

Warm-Up • 1-12 • 56 | 7 | 14 | 10 | 0.5 | 90 | 54 | 8 | 23 | 10 | 1 | 180

Core • 13-22 • 912 | 103 | $\frac{11}{12}$ | 9 | 7 | 1081 | 91 | $\frac{11}{12}$ | 9 | 8

Challenge • 23-26 • 25 | 7 | 15 | 7

Fri • Mission 08 Test

OUT OF 20

Warm-Up • 1-4 • 15 | 10 | 25 | 18

Core • 5-20 • 162 | 71 | 43 | 6 | 43 | 24 | 45 | 12 | 243 | 67 | 44 | 7 | 46 | 29 | 55 | 12

Challenge • 21-24 • 7 | 43 | 7 | 49